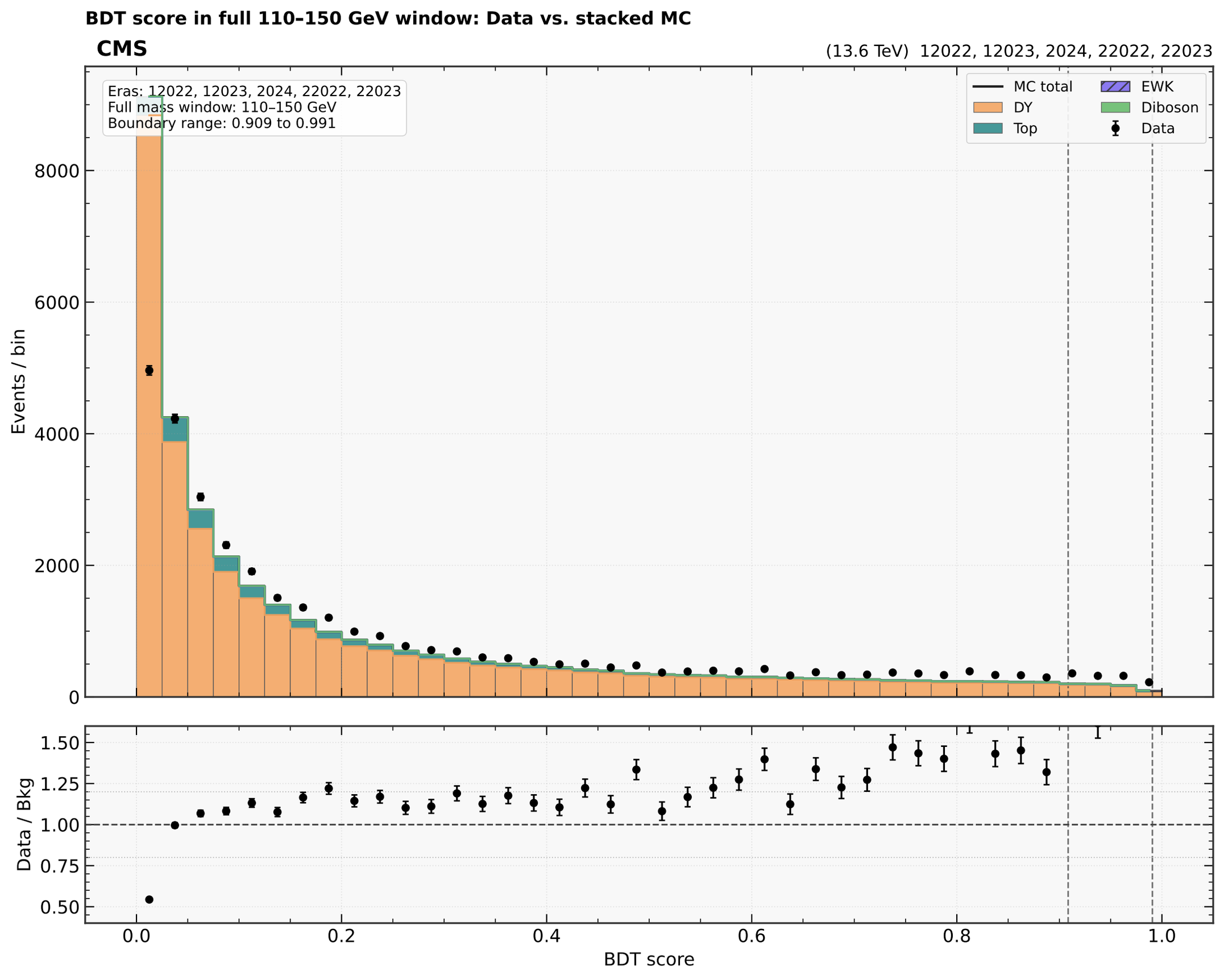
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 12022, 12023, 2024, 22022, 22023
Category edges : 0.0000, 0.9085, 0.9846, 0.9907, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Atanh BDT + S/√B per category
p.6 Signal FWHM vs. BDT score
p.7 ROC curve with AUC
p.8 Blinded dimuon mass spectrum
p.9 Per-category mass distributions and background fits
p.10 Background model comparison (χ^2/ndf)
p.11 VBF signal DCB fits per category
p.12 DCB parameter table
p.13 σ_{CB} vs. category
p.14 Signal acceptance × efficiency
p.15 Per-category yield and sensitivity summary
p.16 Category summary table
p.17 Category process composition

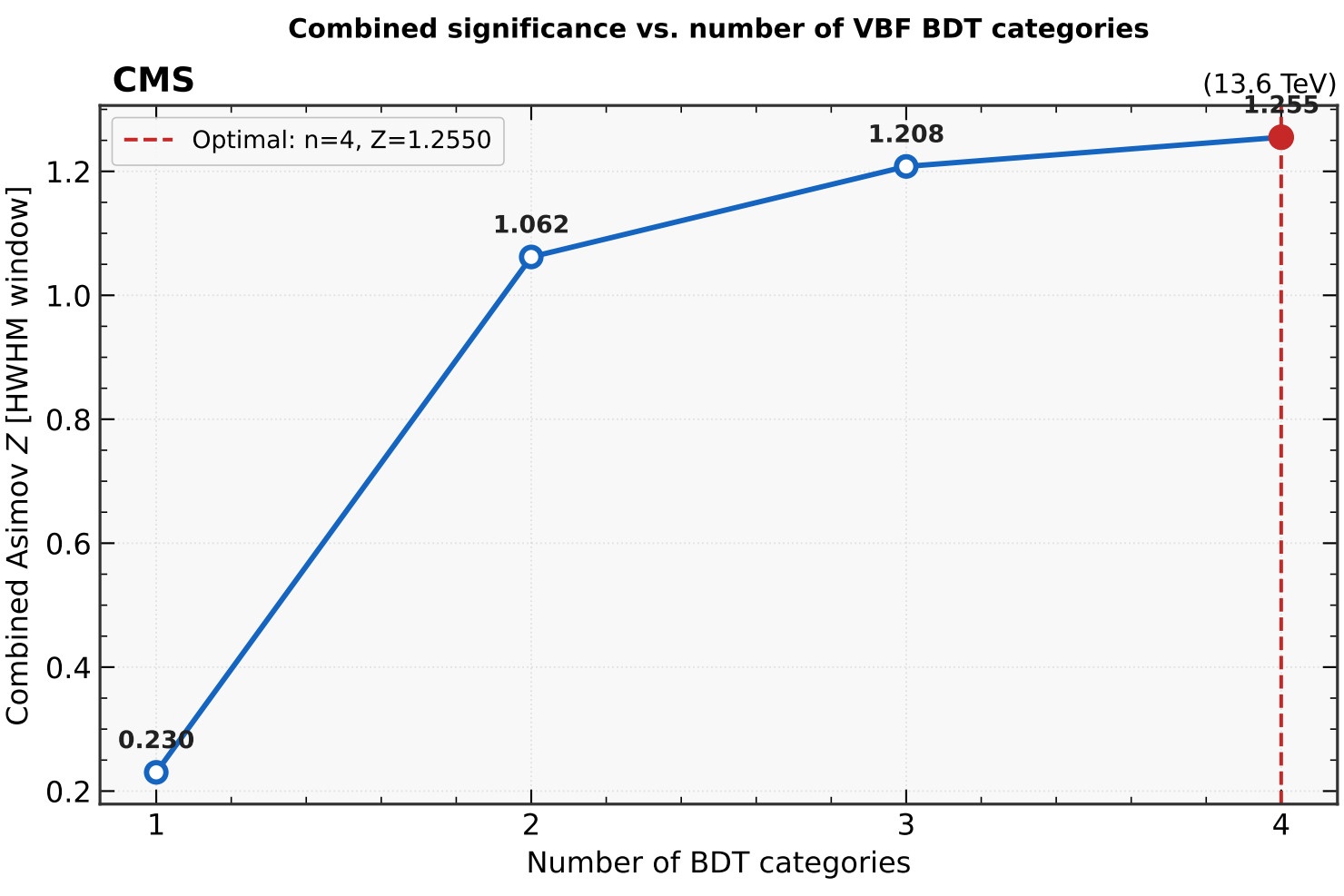

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 12022, 12023, 2024, 22022, 22023
Eras: 12022, 12023, 2024, 22022, 22023
Full mass window: 110–150 GeV
Boundary range: 0.909 to 0.991
MC total
DY
Top
EWK
Diboson
Data
Events / bin
8000
6000
4000
2000
0
Data / Bkg
1.50
1.25
1.00
0.75
0.50
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

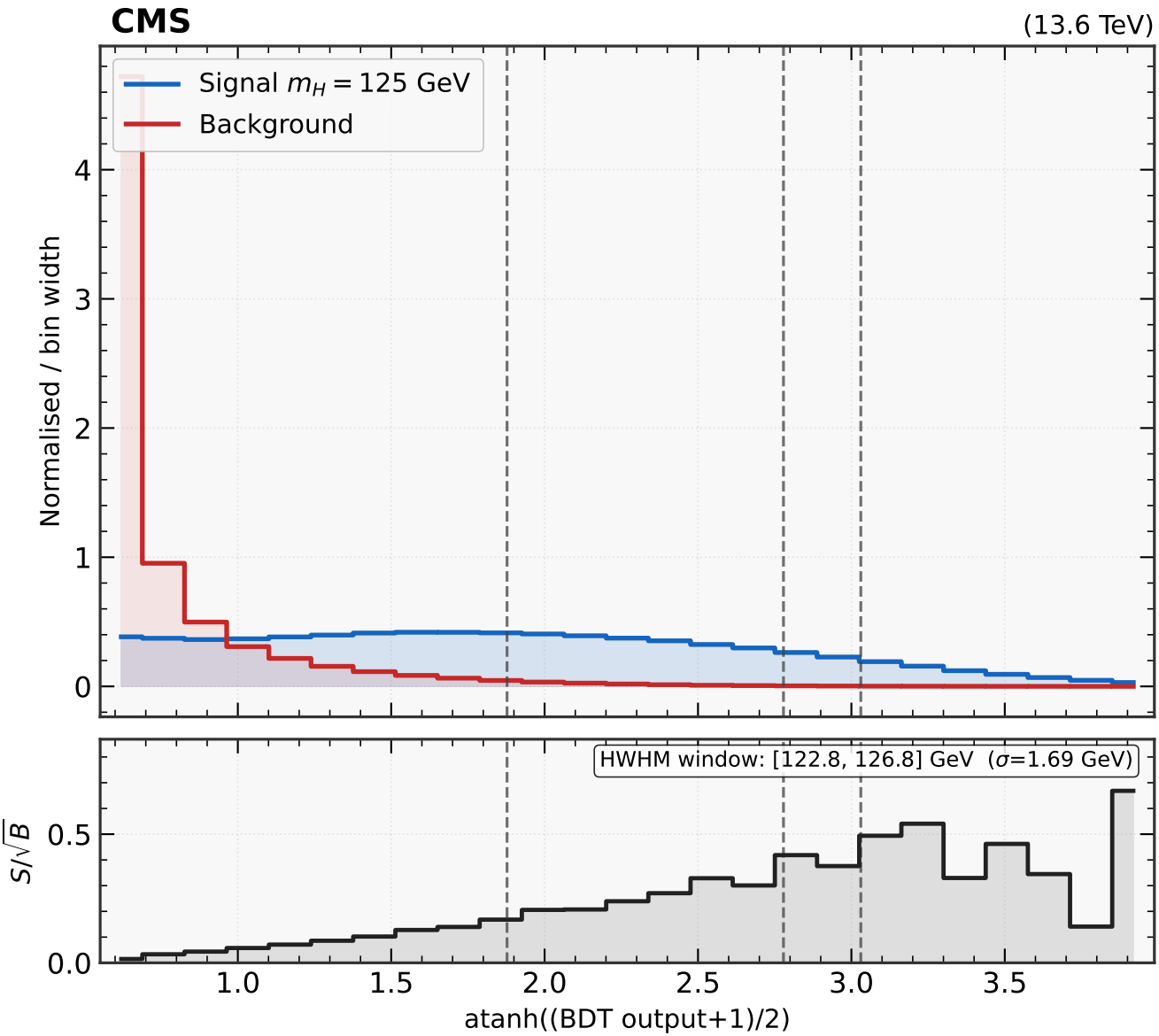
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.245	9125.66 ± 27.49	4962	0.00
[0.025, 0.050)	0.391	4247.18 ± 19.27	4230	0.01
[0.050, 0.075)	0.441	2845.06 ± 15.69	3039	0.02
[0.075, 0.100)	0.451	2132.17 ± 13.58	2308	0.02
[0.100, 0.125)	0.461	1685.71 ± 12.07	1908	0.03
[0.125, 0.150)	0.458	1398.98 ± 11.03	1506	0.03
[0.150, 0.175)	0.460	1167.46 ± 10.08	1360	0.04
[0.175, 0.200)	0.458	987.61 ± 9.22	1205	0.05
[0.200, 0.225)	0.469	867.61 ± 8.68	993	0.05
[0.225, 0.250)	0.478	791.58 ± 8.41	926	0.06
[0.250, 0.275)	0.474	701.36 ± 7.89	773	0.07
[0.275, 0.300)	0.485	640.90 ± 7.52	712	0.08
[0.300, 0.325)	0.497	581.47 ± 7.15	692	0.09
[0.325, 0.350)	0.503	533.76 ± 6.83	601	0.09
[0.350, 0.375)	0.525	500.66 ± 6.71	589	0.10
[0.375, 0.400)	0.537	470.93 ± 6.51	533	0.11
[0.400, 0.425)	0.558	448.80 ± 6.36	496	0.12
[0.425, 0.450)	0.580	413.84 ± 6.10	506	0.14
[0.450, 0.475)	0.603	398.80 ± 6.04	448	0.15
[0.475, 0.500)	0.627	358.83 ± 5.65	479	0.17

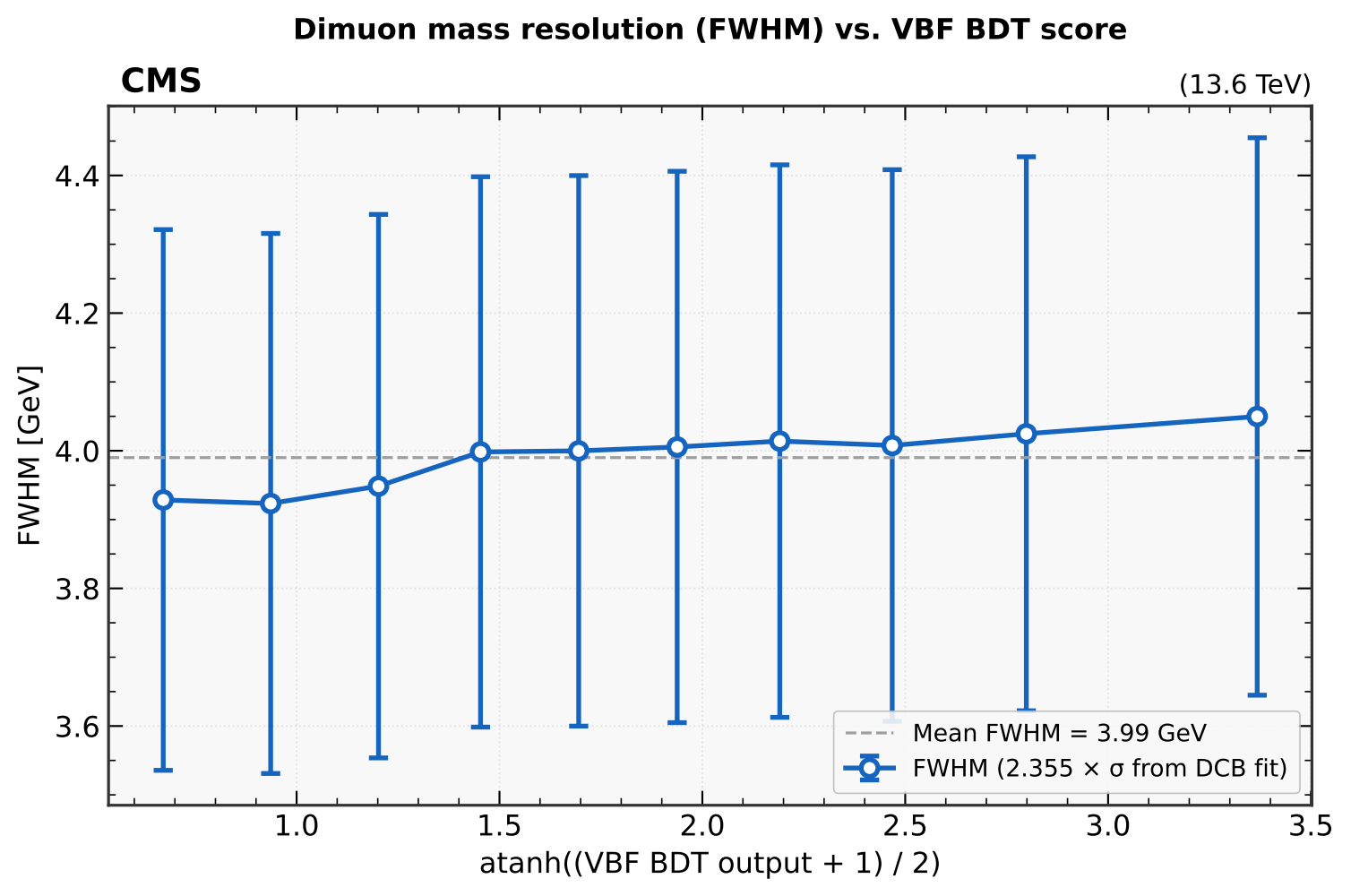
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

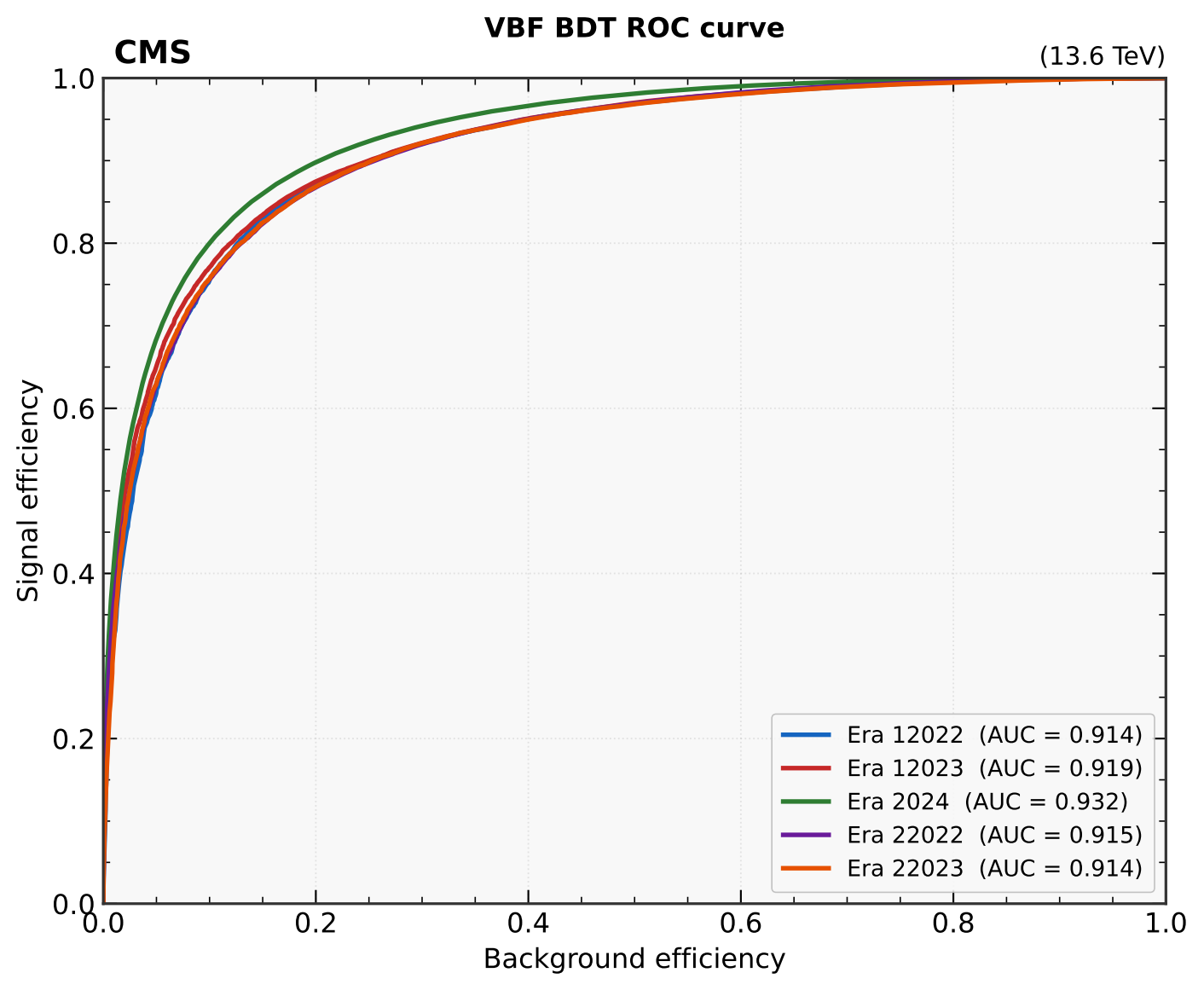
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.652	342.94 ± 5.50	371	0.19
[0.525, 0.550)	0.694	331.33 ± 5.40	387	0.21
[0.550, 0.575)	0.728	325.90 ± 5.40	399	0.22
[0.575, 0.600)	0.782	305.23 ± 5.15	389	0.26
[0.600, 0.625)	0.836	304.06 ± 5.22	425	0.27
[0.625, 0.650)	0.894	290.87 ± 5.09	327	0.31
[0.650, 0.675)	0.958	281.02 ± 5.00	376	0.34
[0.675, 0.700)	1.043	270.74 ± 4.91	332	0.38
[0.700, 0.725)	1.141	267.11 ± 4.83	340	0.43
[0.725, 0.750)	1.262	252.30 ± 4.74	371	0.50
[0.750, 0.775)	1.397	248.87 ± 4.66	357	0.56
[0.775, 0.800)	1.586	237.69 ± 4.55	333	0.66
[0.800, 0.825)	1.800	237.66 ± 4.55	390	0.75
[0.825, 0.850)	2.085	233.32 ± 4.52	334	0.89
[0.850, 0.875)	2.440	227.30 ± 4.44	330	1.06
[0.875, 0.900)	2.955	225.06 ± 4.48	297	1.30
[0.900, 0.925)	3.735	200.72 ± 4.15	359	1.83
[0.925, 0.950)	5.043	198.53 ± 4.15	321	2.48
[0.950, 0.975)	7.753	176.94 ± 3.92	321	4.20
[0.975, 1.000)	14.549	99.22 ± 2.79	224	12.79

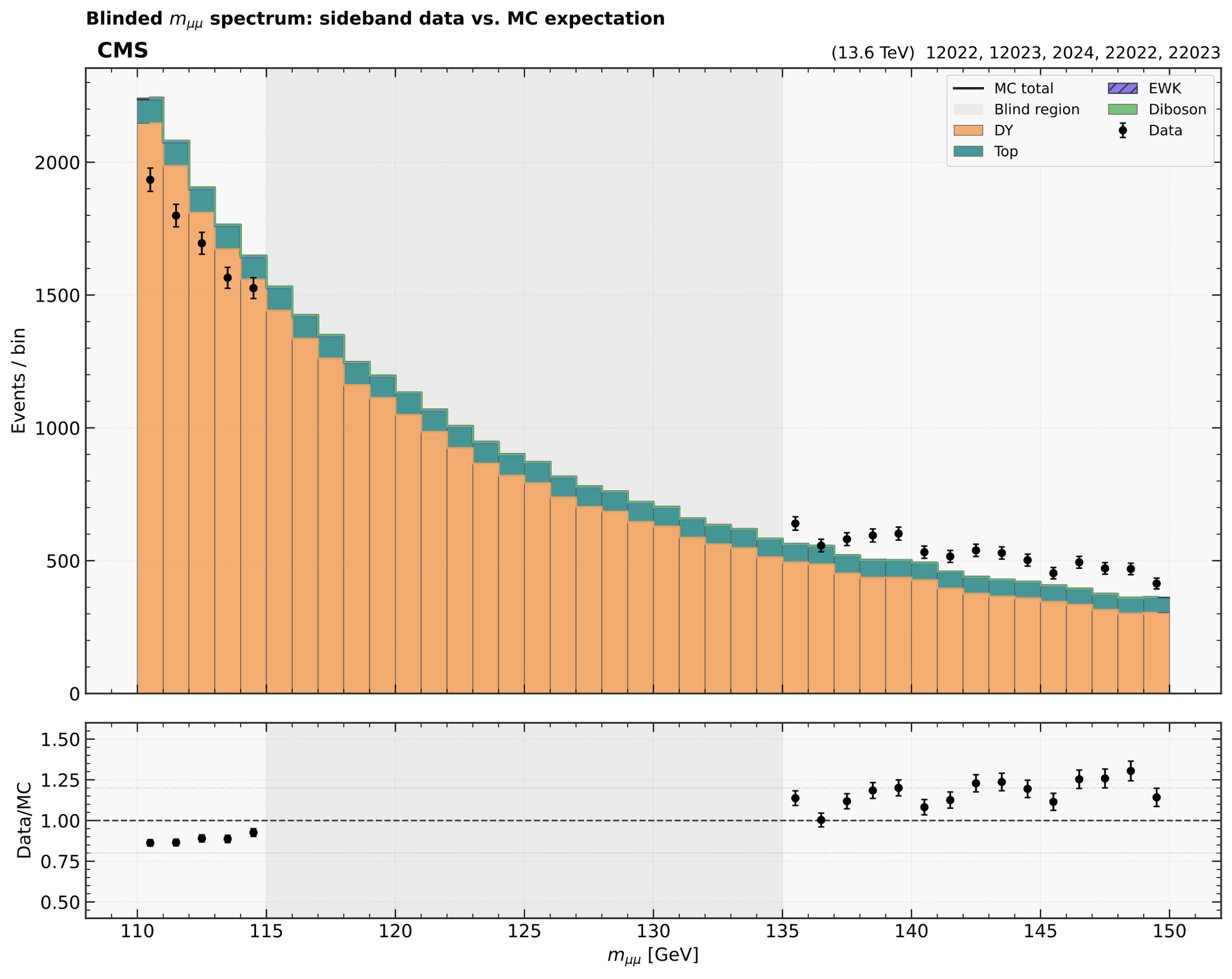

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=1.2550
0.230
1.062
1.208
1.255
Combined Asimov Z [HWHM window]
Number of BDT categories
0.2
0.4
0.6
0.8
1.0
1.2
1
2
3
4

VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window

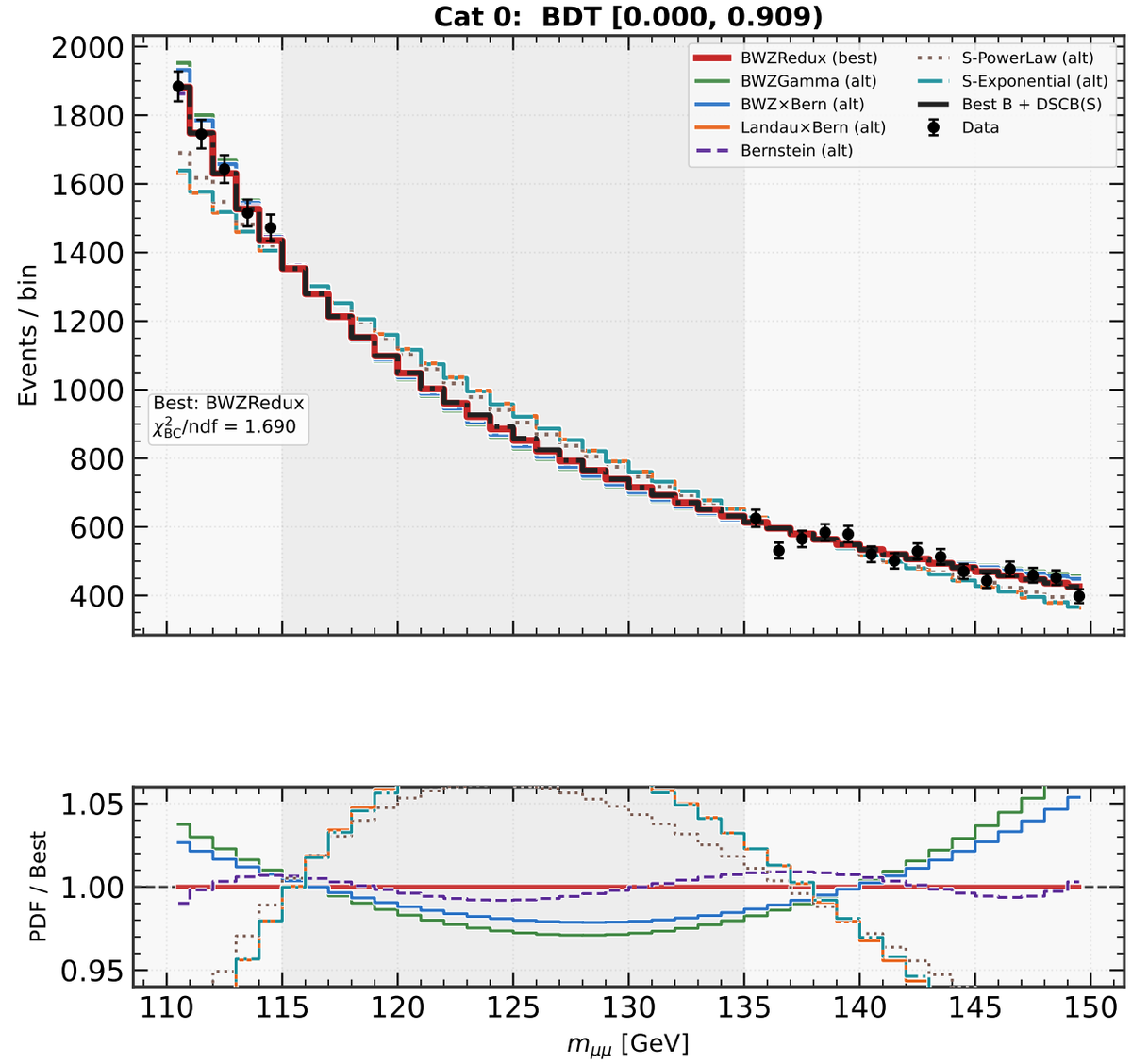


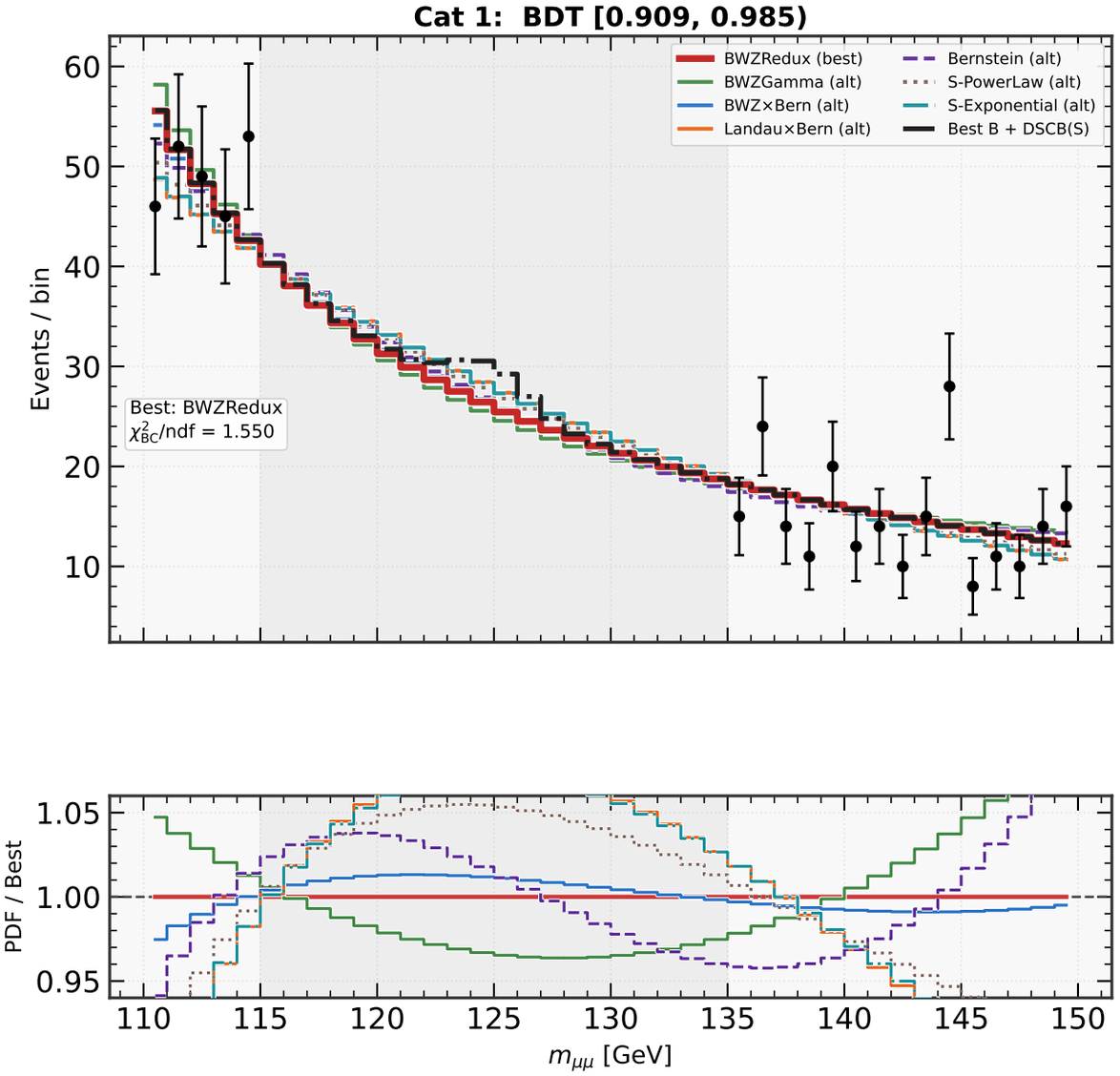

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
1.0
1.5
2.0
2.5
3.0
3.5
atanh((VBF BDT output + 1) / 2)
Mean FWHM = 3.99 GeV
FWHM (2.355 × σ from DCB fit)

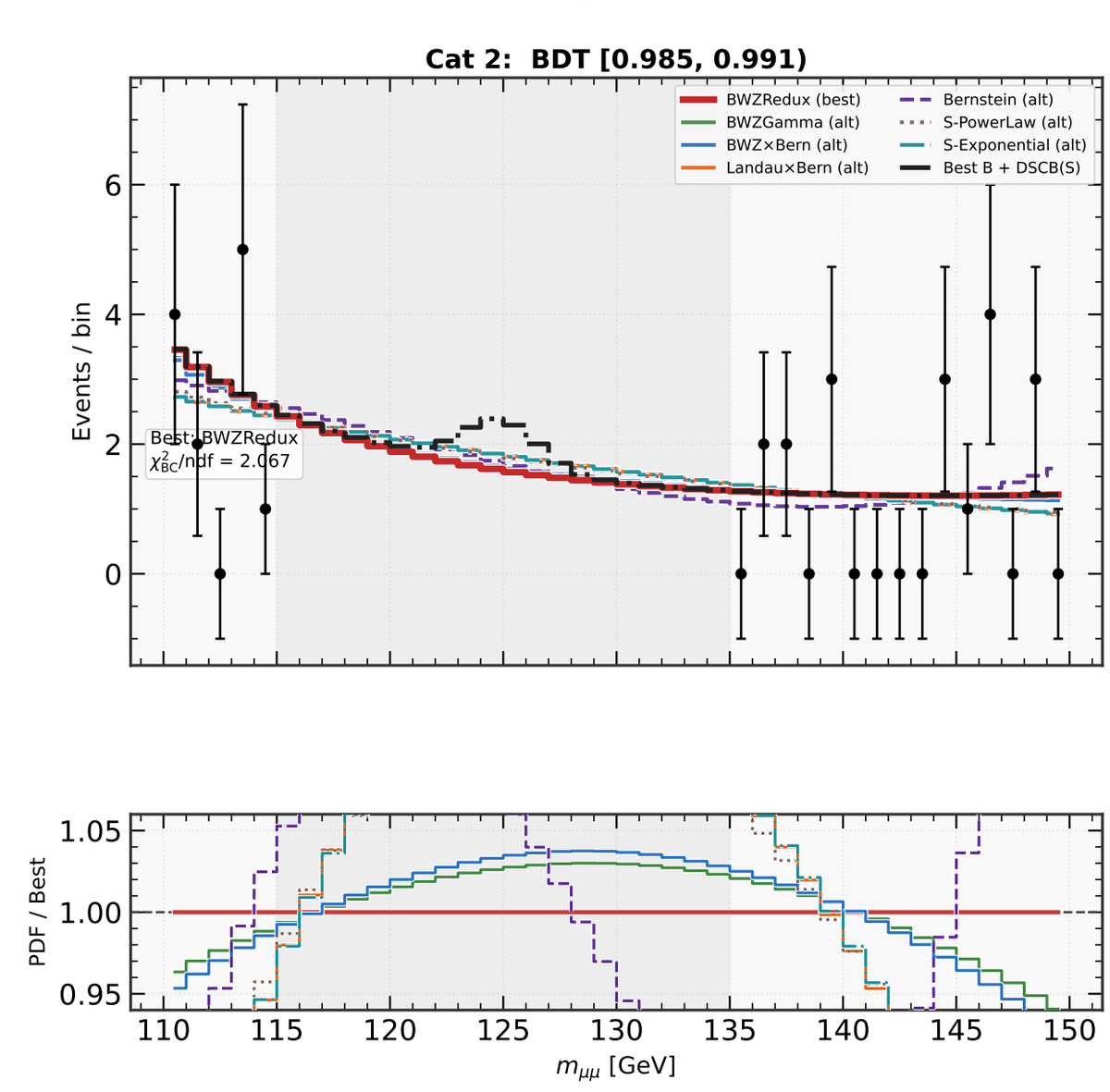

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 12022 (AUC = 0.914)
Era 12023 (AUC = 0.919)
Era 2024 (AUC = 0.932)
Era 22022 (AUC = 0.915)
Era 22023 (AUC = 0.914)

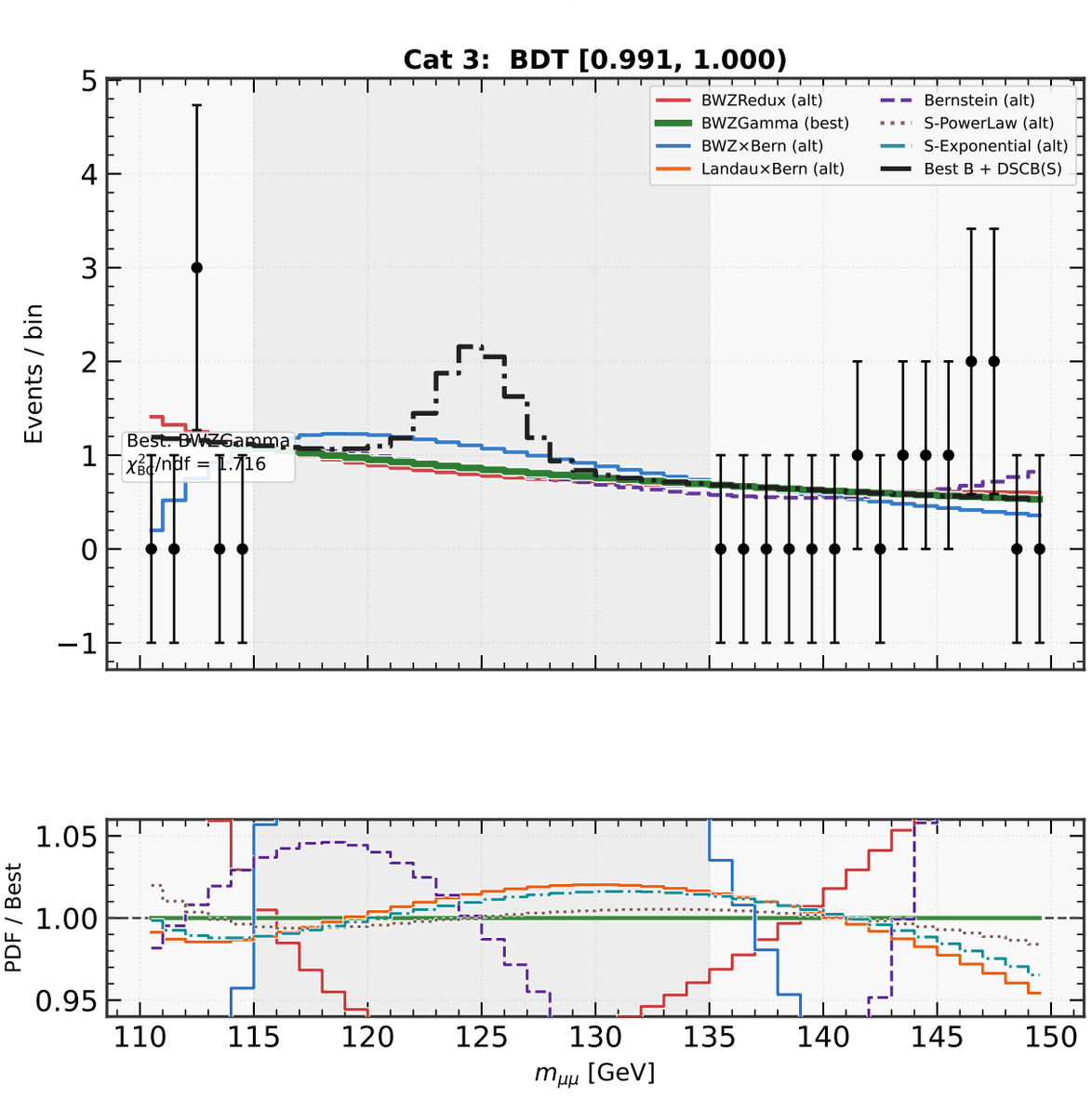

Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 12022, 12023, 2024, 22022, 22023
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits





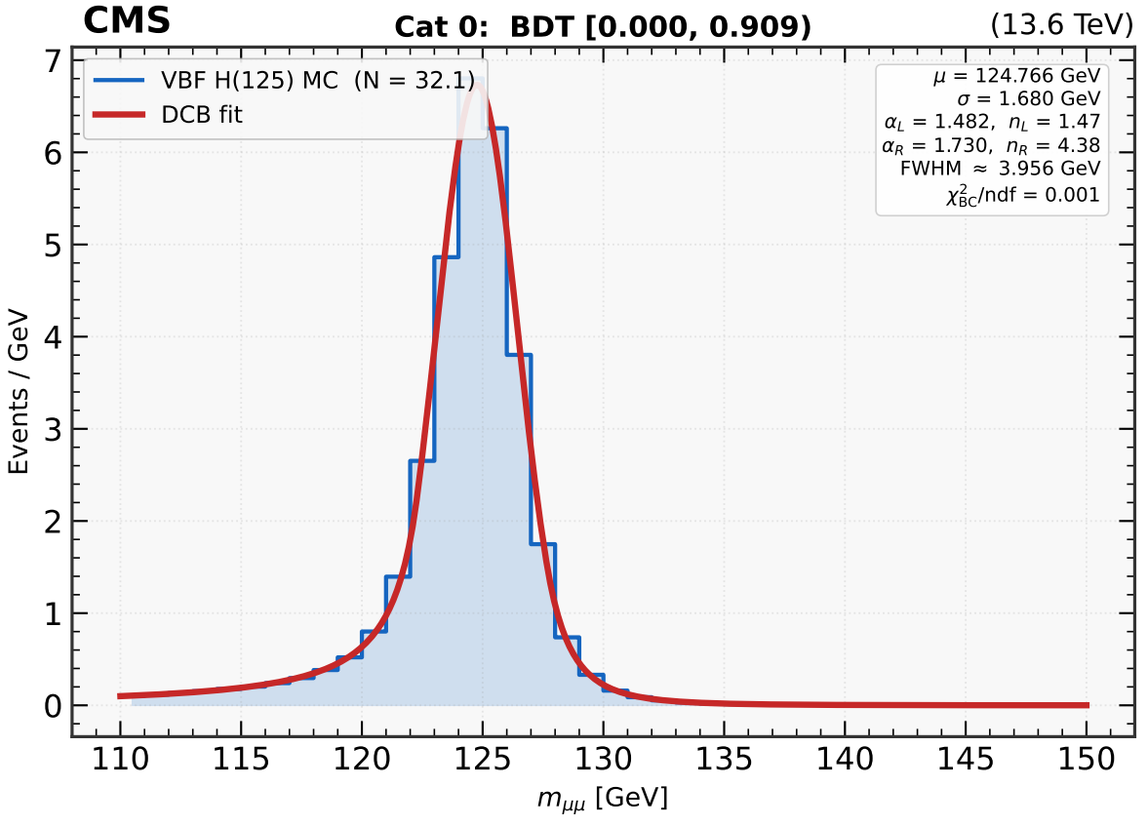


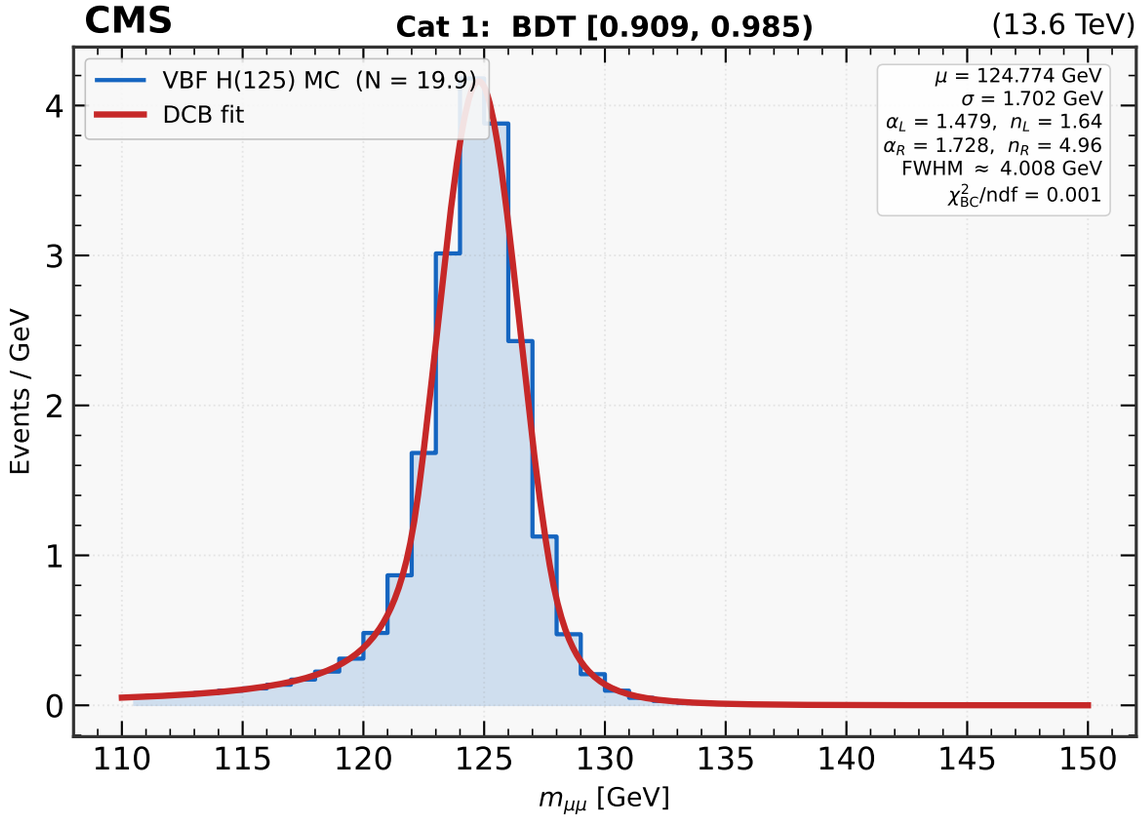


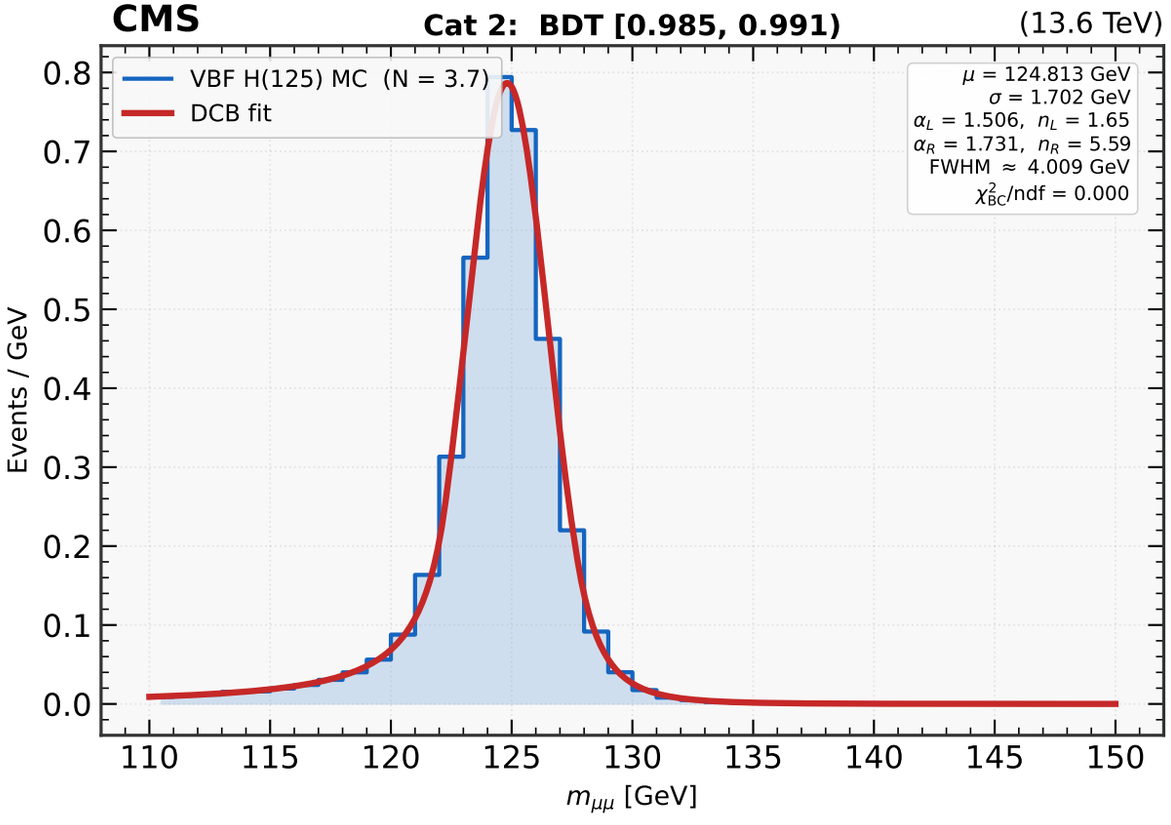
Per-category background model fit comparison (χ²_BC/ndf, ndf)

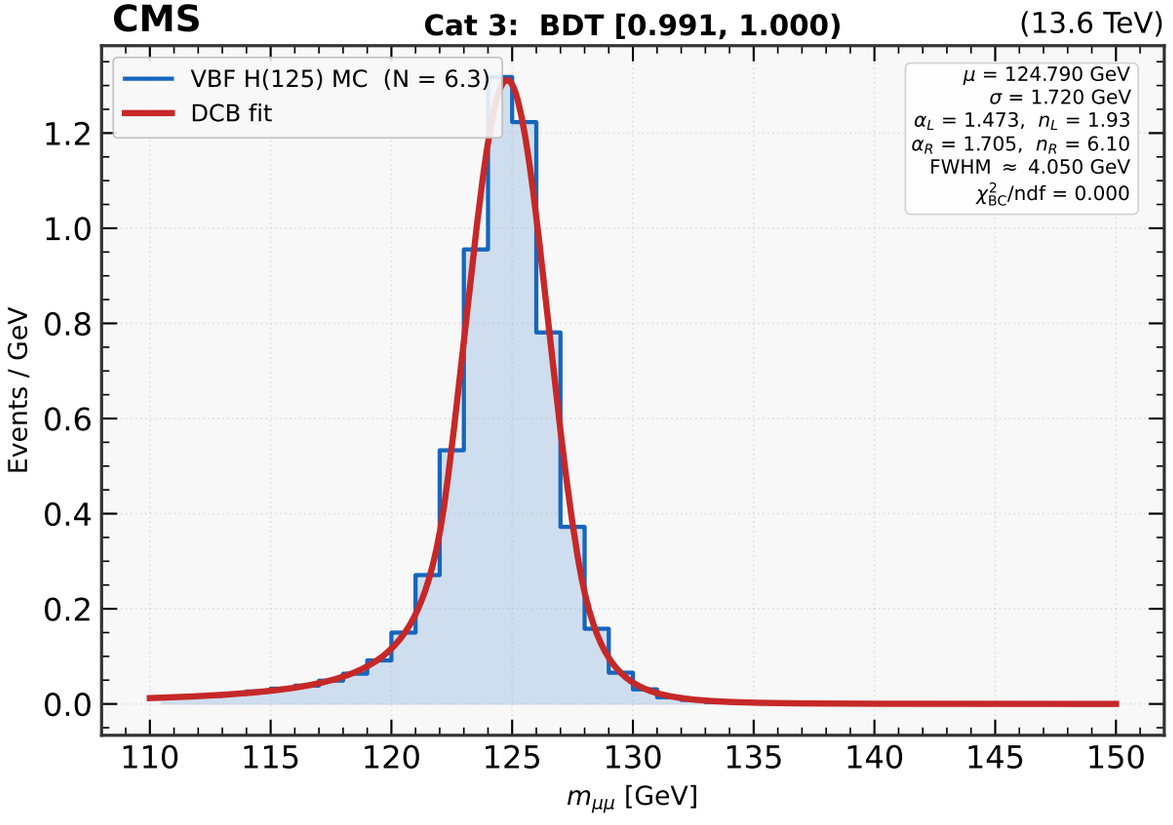
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.909)	BWZRedux	1.690	1.690 (37)	2.267 (37)	2.057 (36)	6.567 (37)	1.909 (34)	4.376 (37)	6.280 (37)
1	[0.909, 0.985)	BWZRedux	1.550	1.550 (37)	1.577 (37)	1.582 (36)	1.632 (37)	1.605 (35)	1.577 (37)	1.626 (37)
2	[0.985, 0.991)	BWZRedux	2.067	2.067 (37)	2.069 (37)	2.127 (36)	2.085 (37)	2.134 (35)	2.079 (37)	2.084 (37)
3	[0.991, 1.000)	BWZGamma	1.716	1.729 (37)	1.716 (37)	1.730 (36)	1.721 (37)	1.752 (35)	1.720 (37)	1.721 (37)

VBF H(125) signal DCB extraction per BDT category



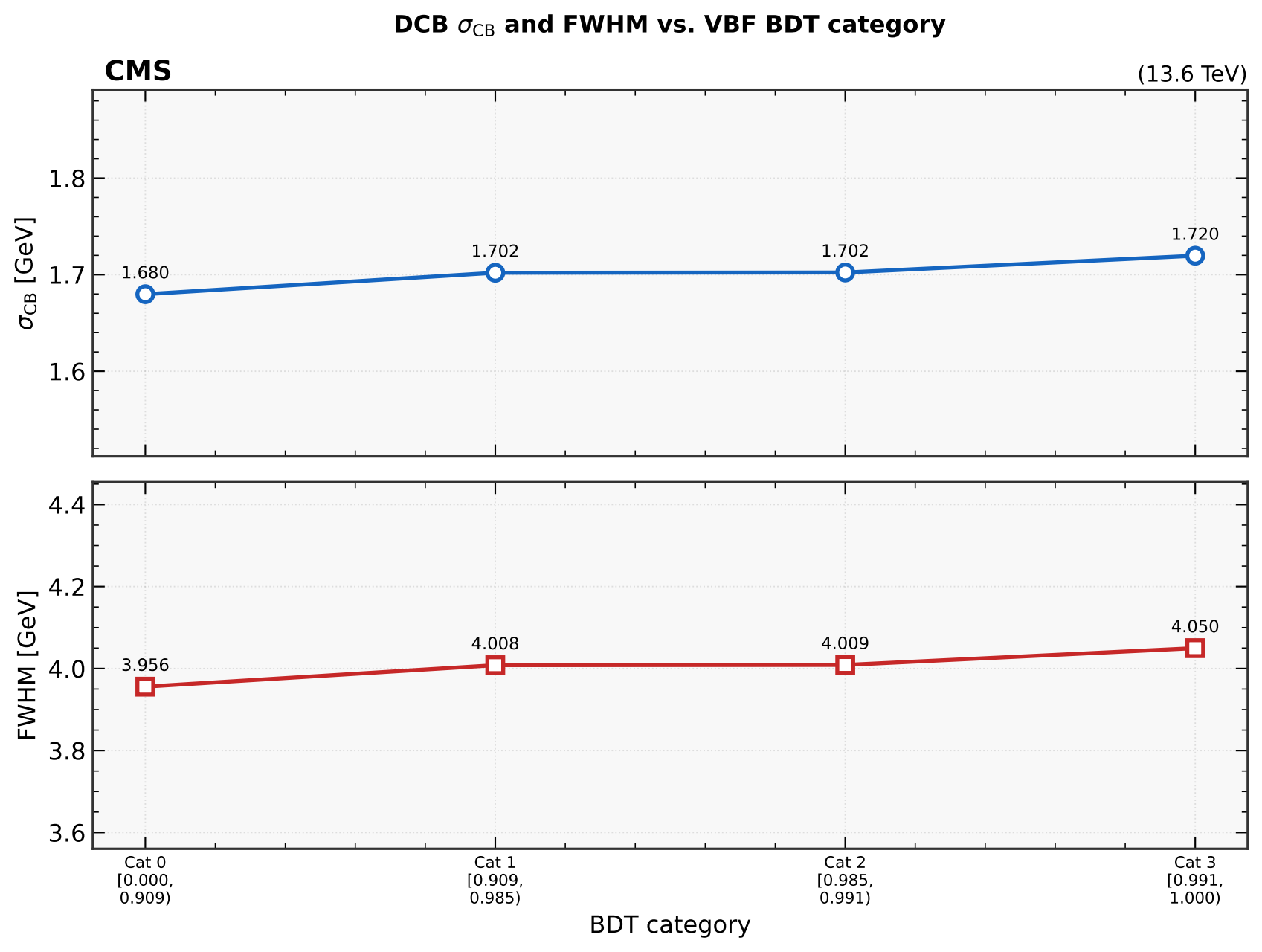


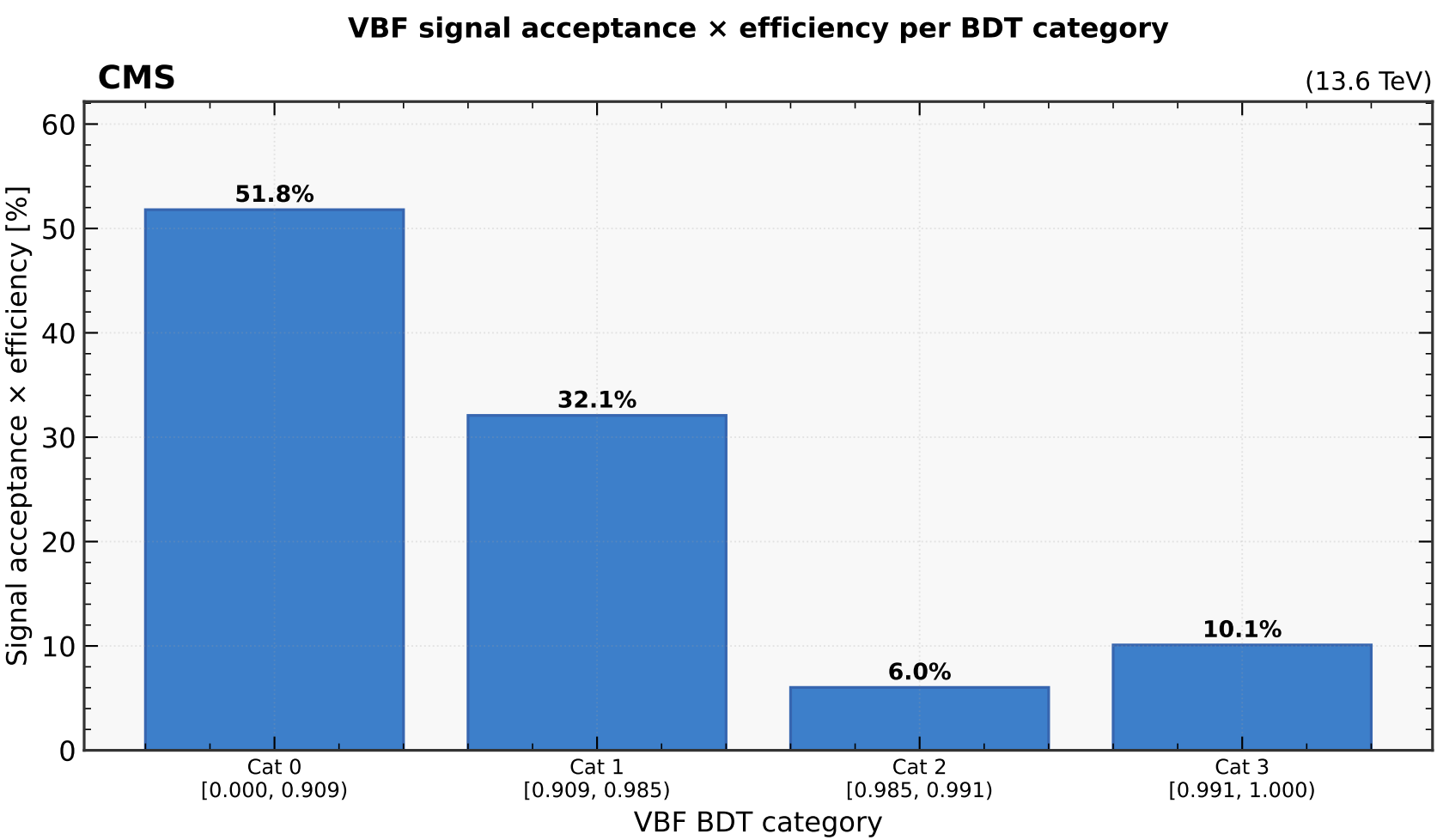




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.909)	32.13	124.7655	1.6798	1.4818	1.467	1.7301	4.382	3.956	0.120	0.001
1	[0.909, 0.985)	19.90	124.7744	1.7020	1.4793	1.640	1.7279	4.961	4.008	0.596	0.001
2	[0.985, 0.991)	3.74	124.8134	1.7023	1.5064	1.646	1.7307	5.588	4.009	0.526	0.000
3	[0.991, 1.000)	6.26	124.7904	1.7197	1.4731	1.925	1.7052	6.099	4.050	0.967	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.8
1.7
1.6
1.680
1.702
1.702
1.720
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
3.956
4.008
4.009
4.050
Cat 0 [0.000, 0.909)
Cat 1 [0.909, 0.985)
Cat 2 [0.985, 0.991)
Cat 3 [0.991, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
60
51.8%
32.1%
6.0%
10.1%
Cat 0
[0.000, 0.909)
Cat 1
[0.909, 0.985)
Cat 2
[0.985, 0.991)
Cat 3
[0.991, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.909)	32.13	327559.74	21.710	32802.918	[122.8, 126.7]	0.120	0.120
1	[0.909, 0.985)	19.90	5276.83	13.584	515.644	[122.8, 126.8]	0.598	0.596
2	[0.985, 0.991)	3.74	259.96	2.564	22.923	[122.8, 126.8]	0.536	0.526
3	[0.991, 1.000)	6.26	175.46	4.329	18.648	[122.8, 126.8]	1.002	0.967

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.909)	15905	152100.50 ± 351.25	0.105	327559.74 ± 516.28	0.1199	0.1199	32802.918
1	[0.909, 0.985)	467	2414.89 ± 44.30	0.193	5276.83 ± 65.60	0.5982	0.5956	515.644
2	[0.985, 0.991)	30	120.54 ± 9.10	0.249	259.96 ± 13.68	0.5355	0.5260	22.923
3	[0.991, 1.000)	11	79.55 ± 6.78	0.138	175.46 ± 10.18	1.0023	0.9669	18.648
Combined	all						1.2572	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total	total_err2
0	[0.000, 0.909)	299942.35	422.45	26624.56	570.39	327559.74	266548.03
1	[0.909, 0.985)	4672.36	213.48	373.67	17.31	5276.83	4303.15
2	[0.985, 0.991)	208.16	36.29	14.79	0.73	259.96	187.16
3	[0.991, 1.000)	120.02	45.82	9.28	0.34	175.46	103.63